Calculus II Assignment 7

Part I. Regular Problems

1. Use equations and inequalities to describe the following sets in polar coordinates.

   (a) The open unit disk $x^2 + y^2 < 1$.

   (b) The first quadrant $x > 0, y > 0$.

2. Match the formulas (i-iv) in spherical coordinates with the descriptions (a-d) of sets in $\mathbb{R}^3$. Note that if we don't specify any restrictions on a coordinate, then the usual ones apply: $\rho \geq 0, 0 \leq \phi \leq \pi, 0 \leq \theta \leq 2\pi$.

$$(i)\ \rho < 1 \quad (ii)\ \phi = \frac{2\pi}{3} \quad (iii)\ 0 < \theta < \pi \quad (iv)\ \rho > 1 \text{ and } \frac{\pi}{2} < \phi$$

   (a) a half-space
   (b) an open ball
   (c) a cone surface
   (d) a half-space with points removed that were within one unit of the origin.

3. Find the linear approximation of the function $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ at $(3, 2, 6)$ and use it to approximate the number $\sqrt{(3.02)^2 + (1.97)^2 + (5.99)^2}$.

4. Compute $\partial f/\partial x$ and $\partial f/\partial y$ for the functions in (a) and (b), and compute $\partial z/\partial x$ and $\partial z/\partial y$ in (c) and (d); evaluate at the indicated points.

   (a) $f(x, y) = xy; (1, 1)$

   (b) $f(x, y) = \frac{x}{y}; (1, 1)$

   (c) $z = 3x^2 + 2y^2; (1, 2)$

   (d) $z = \sin\langle x^2 - 3xy\rangle; (\sqrt{\pi}, \sqrt{\pi})$

5. Compute f_x, f_y, and f_z for the following functions, and evaluate them at the indicated points.

   (a) $f(x, y, z) = xyz; (1, 1, 1)$

   (b) $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}; (3, 0, 4)$

   (c) $f(x, y, z) = \cos(xy^2) + e^{3xyz}; (\pi, 1, 1)$

   (d) $f(x, y, z) = x^{yz}; (1, 1, 0)$

Part II. Group Problems

1. Find the indicated partial derivatives of the functions.

   (a) $f_x(x, y)$ and $f_y(2, 0)$ if $f(x, y) = \mathrm{e}^{-x^2-y^2}$

   (b) $\frac{\partial}{\partial y}\left(x\mathrm{e}^y + y\mathrm{e}^x\right)$

   (c) $\frac{\partial}{\partial x}\left(\cos(xy) + \frac{\partial}{\partial y}(\sin(xy))\right)$

2. Let $f(x, y) = x^2y^3$. Let $(x, y) = (a + u, b + v)$. Use binomial expansion on the right side of

   $$x^2y^3 = (a + u)^2(b + v)^3$$

   to find the numbers c_1, c_2, c_3 in

   $$f(x, y) = c_1 + c_2(x - a) + c_3(y - b) + \cdots$$

   where the dots represent polynomials in $(x - a)$ and $(y - b)$ of degree 2 or more.

   (a) Express c_1 in terms of f, a, and b.

   (b) Express c_2 and c_3 in terms of partial derivatives of f at (a, b).

   (c) Find functions ℓ and s so that

   $$f(a + u, b + v) = f(a, b) + \ell(u, v) + s(u, v)$$

   and ℓ is linear, and $s(u, v)$ is small compared to (u, v) as (u, v) tends to $(0, 0)$.